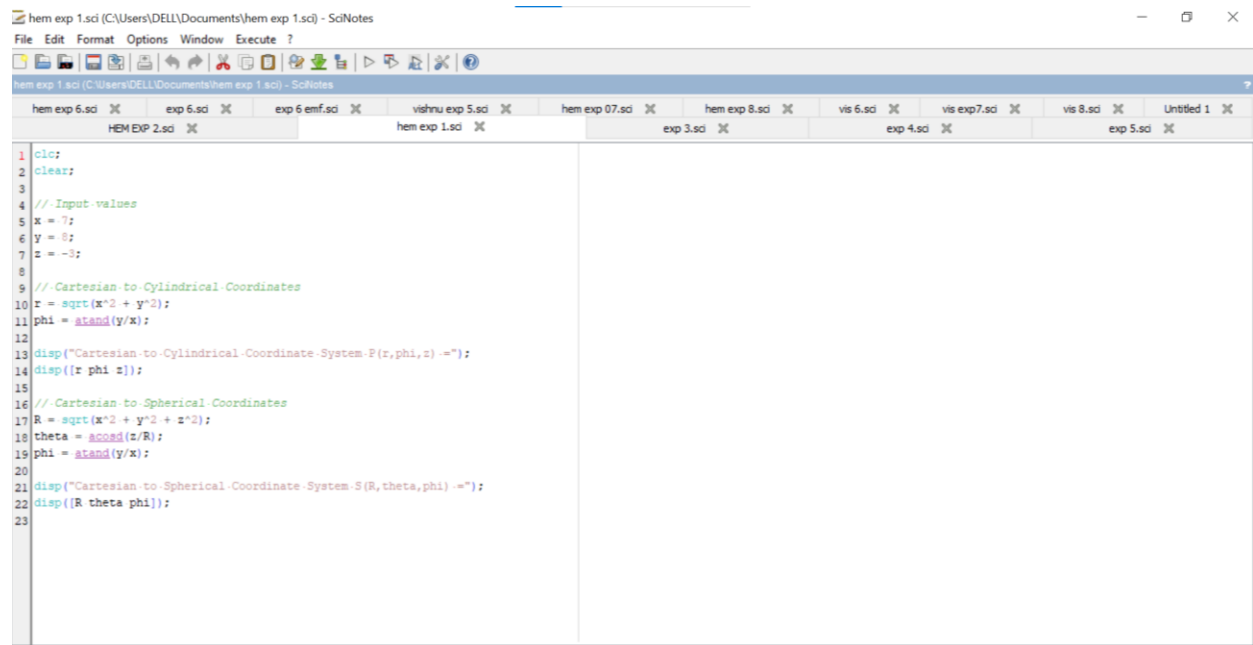


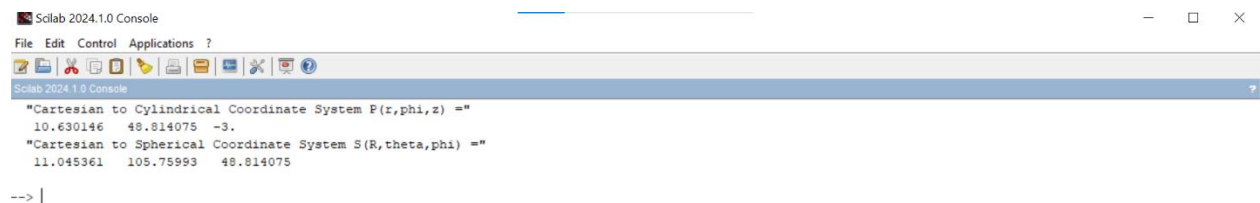
EXP 1: Transformation between Cartesian, Cylindrical and spherical coordinate system

Test case (i) Transformation of Cartesian coordinate points in to Cylindrical and spherical coordinate system



The screenshot shows a Scilab editor window titled "hem exp 1.sci (C:\Users\DELL\Documents\hem exp 1.sci) - SciNotes". The window contains the following code:

```
1 clc;
2 clear;
3
4 // Input values
5 x = 7;
6 y = 8;
7 z = -3;
8
9 // Cartesian to Cylindrical Coordinates
10 r = sqrt(x^2 + y^2);
11 phi = atan2(y/x);
12
13 disp("Cartesian to Cylindrical Coordinate System P(r,phi,z) =");
14 disp([r phi z]);
15
16 // Cartesian to Spherical Coordinates
17 R = sqrt(x^2 + y^2 + z^2);
18 theta = acosd(z/R);
19 phi = atan2(y/x);
20
21 disp("Cartesian to Spherical Coordinate System S(R,theta,phi) =");
22 disp([R theta phi]);
23
```



The screenshot shows a Scilab console window titled "Scilab 2024.1.0 Console". The window displays the output of the code executed in the editor above:

```
"Cartesian to Cylindrical Coordinate System P(r,phi,z) ="
10.630146  48.814075  -3.
"Cartesian to Spherical Coordinate System S(R,theta,phi) ="
11.045361  105.75993  48.814075

--> |
```

Qp:-1

carte(r)

Rectangle to cylindrical

~~$x=7$~~
 x, y, z

$$r = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1}(y/x)$$

$$z = z$$

$$x = 7$$

$$y = 8$$

$$z = -3$$

$$x = 7$$

$$y = 8$$

$$z = -3$$

$$r = \sqrt{x^2 + y^2}$$

$$= \sqrt{7^2 + 8^2}$$

$$= \sqrt{49 + 64}$$

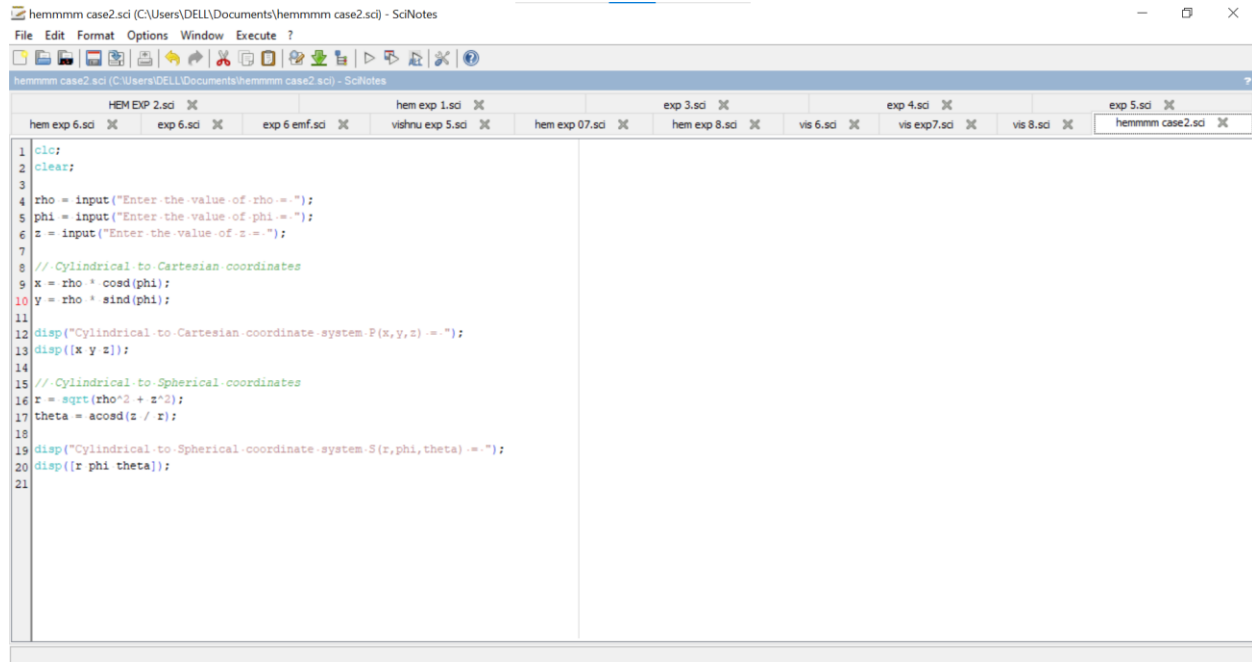
$$= \sqrt{113} = 10.6301$$

$$\phi = \tan^{-1}\left(\frac{8}{7}\right)$$

$$= 48.8141^\circ$$

$$z = -3$$

Test case (ii) Transformation of Cylindrical coordinate points into Cartesian and spherical coordinate system

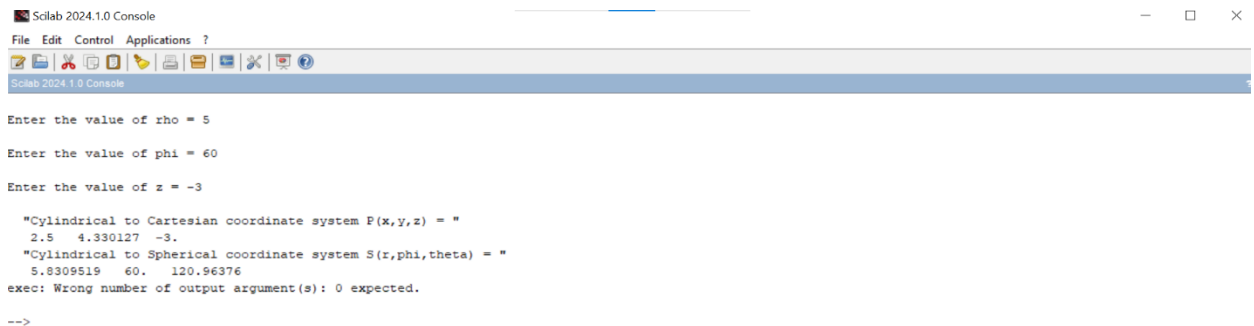


hemmmm case2.sci (C:\Users\DELL\Documents\hemmmm case2.sci) - SciNotes

File Edit Format Options Window Execute ?

hem exp 6.sci exp 6.sci exp 6 emf.sci hem exp 1.sci exp 3.sci exp 4.sci exp 5.sci

```
1 clc;
2 clear;
3
4 rho = input("Enter the value of rho = ");
5 phi = input("Enter the value of phi = ");
6 z = input("Enter the value of z = ");
7
8 // Cylindrical to Cartesian coordinates
9 x = rho * cosd(phi);
10 y = rho * sind(phi);
11
12 disp("Cylindrical to Cartesian coordinate system P(x,y,z) = ");
13 disp([x y z]);
14
15 // Cylindrical to Spherical coordinates
16 r = sqrt(rho^2 + z^2);
17 theta = acosd(z / r);
18
19 disp("Cylindrical to Spherical coordinate system S(r,phi,theta) = ");
20 disp([r phi theta]);
21
```



Scilab 2024.1.0 Console

File Edit Control Applications ?

```
Enter the value of rho = 5
Enter the value of phi = 60
Enter the value of z = -3

"Cylindrical to Cartesian coordinate system P(x,y,z) = "
2.5  4.330127  -3.
"Cylindrical to Spherical coordinate system S(r,phi,theta) = "
5.8309519  60.  120.96376
exec: Wrong number of output argument(s): 0 expected.

-->
```

$$z = 7$$

case-2

cylindrical to spherical.

$$\rho = 5$$

$$\phi = 60^\circ$$

$$z = -3$$

$$x = \sqrt{\rho^2 + z^2}$$

$$= \sqrt{5^2 + (-3)^2}$$

$$= \sqrt{25 + 9} = \sqrt{34}$$

$$\Rightarrow 5.8310$$

$$y = \cos^{-1} \left(\frac{z}{\rho} \right)$$

$$= \cos^{-1} \left(\frac{-3}{5} \right)$$

$$= \cos^{-1} (-0.6)$$

$$= 120.96^\circ$$

$$z = 60^\circ$$

$$x = \rho \cos \theta$$

$$y = \rho \sin \theta$$

$$z = z$$

$$x = 5 \cos 60^\circ$$

$$= 5 \times 0.5$$

$$= 2.5$$

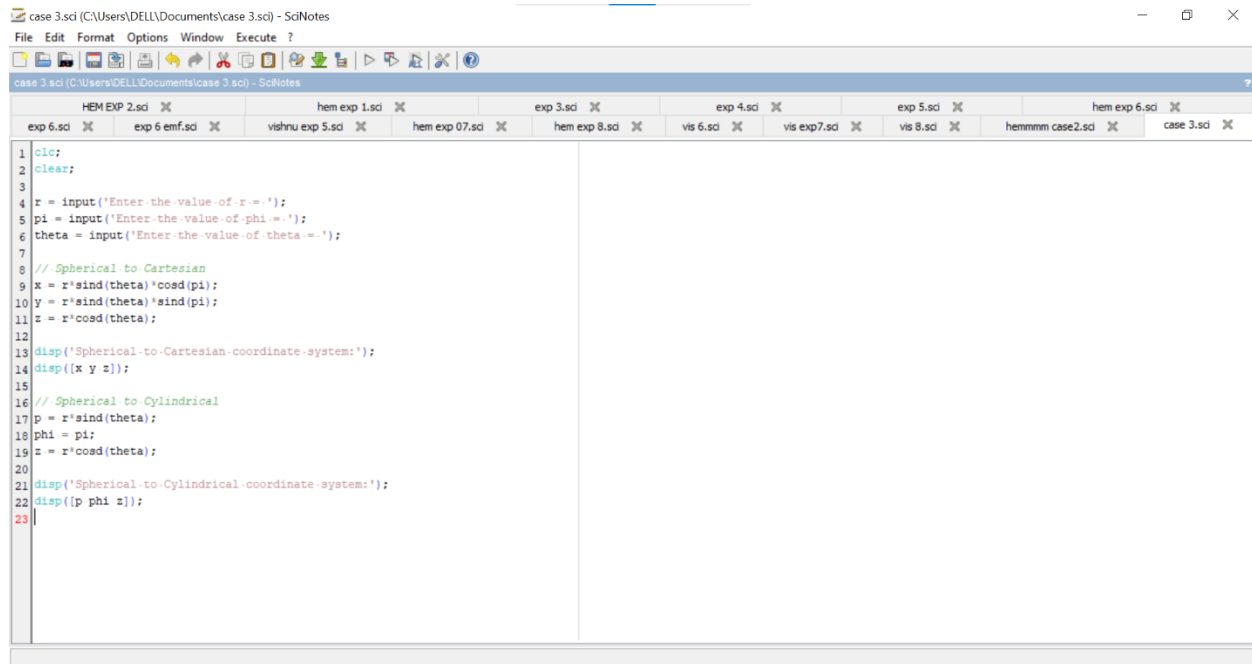
$$y = 5 \sin 60^\circ$$

$$= 5 \times 0.8660$$

$$= 4.3301$$

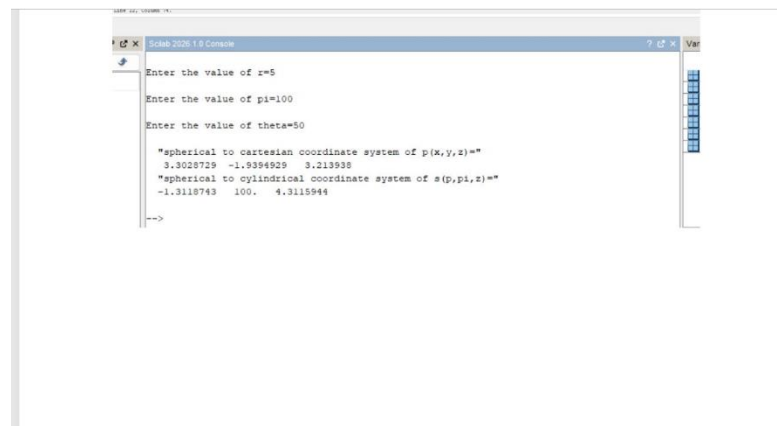
$$z = -3$$

Test case (iii) Transformation of spherical coordinate points into Cylindrical and Cartesian coordinate system



The screenshot shows a MATLAB script editor window titled "case 3.sci (C:\Users\DELL\Documents\case 3.sci) - SciNotes". The script contains the following code:

```
1 clc;
2 clear;
3
4 r = input('Enter the value of r = ');
5 pi = input('Enter the value of phi = ');
6 theta = input('Enter the value of theta = ');
7
8 // Spherical to Cartesian
9 x = r*sind(theta)*cosd(pi);
10 y = r*sind(theta)*sind(pi);
11 z = r*cosd(theta);
12
13 disp('Spherical to Cartesian coordinate system:');
14 disp([x y z]);
15
16 // Spherical to Cylindrical
17 p = r*sind(theta);
18 phi = pi;
19 z = r*cosd(theta);
20
21 disp('Spherical to Cylindrical coordinate system:');
22 disp([p phi z]);
23
```



The screenshot shows a MATLAB console window titled "Sciab 2020 1.0 Console". The console displays the following output:

```
Enter the value of r=5
Enter the value of pi=100
Enter the value of theta=50

"spherical to cartesian coordinate system of p(x,y,z) ="
3.3020729 -1.9394929 3.213938
"spherical to cylindrical coordinate system of s(p,pi,z) ="
-1.3118743 100. 4.3115944
```

Ex 111

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$r = 5$$

$$\theta = 30^\circ$$

$$\phi = 75^\circ$$

$$x = 5 \times \sin 30^\circ \times \cos 75^\circ$$

$$= 5 \times 0.5 \times 0.268$$

$$= 0.647$$

$$y = 5 \times \sin 30^\circ \times \sin 75^\circ$$

$$= 5 \times 0.5 \times 0.9659$$

$$= 2.415$$

$$z = 5 \times \cos 30^\circ$$

$$= 5 \times 0.8660$$

$$= 4.330$$

Spherical to cylindrical:

$$\rho = 5 \times \sin 30^\circ$$

$$= 5 \times 0.5$$

$$= 2.5$$

$$\theta = 75^\circ$$

$$\phi = 5 \times \cos 30^\circ$$

$$= 5 \times 0.8660$$

$$= 4.330$$

$$\rho = r \sin \theta$$

$$\phi = \phi$$

$$z = r \cos \theta$$